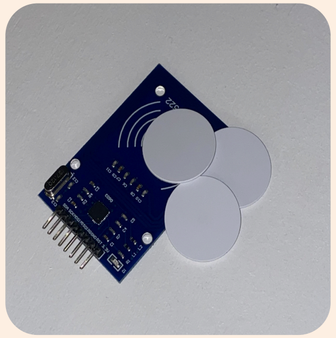
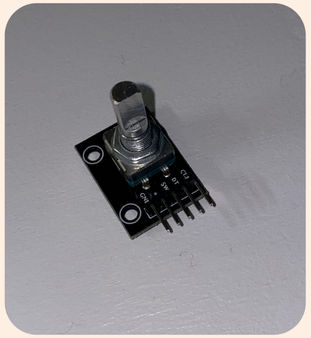
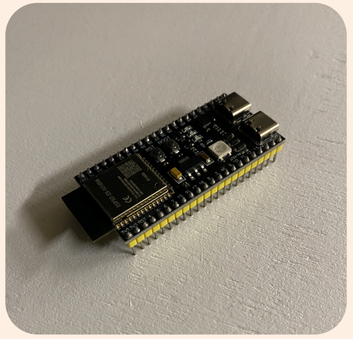
Electronics - Design of the Solution

The tangibles have got NFC-Chips at the bottom. They contain a text information unique to the ant- and the human-tangible. The ant-tangible has an “A” as information on the NFC-Chip, while the human-tangible has an “H”.

Below the desktop are NFC-Readers at each of the five ports that pick up the information written on the NCF-Chip.
The NFC-Readers are connected to the Arduino and send a Signal via cable and Serial Monitor to the Website. It is setup to contain the information of each of the readers if there is a tangible currently placed at the port.

Regarding the registration of turning, we are using five rotary-encoder, which are setup in two different modes:
The ports one and two are setup to track the rotation in degrees and send the information via Serial Monitor or cable to the website. The website turns the 3D-Brains accordingly to the degree registered by the encoders.

The ports three to five are setup to have 4 points of interest at the degrees 0, 90, 180 and 270.
For each of those points there is a certain information to be displayed by the website.